Comparison of VQE, VQITE, and TTITE for Charmonium Spectroscopy

Apples-to-Apples Benchmark on the Woloshyn Hamiltonian

quarkonium-simulations project

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Abstract

We compare three quantum-computing methods—the Variational Quantum Eigensolver (VQE), Variational Quantum Imaginary-Time Evolution (VQITE), and Trotter–Taylor Imaginary-Time Evolution (TTITE)—on the same 2-qubit charmonium Hamiltonian with spin-dependent interactions. All three methods use the same direct qubit encoding and the same physical parameters, enabling a controlled comparison of accuracy, convergence speed, and noise sensitivity.

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1 Setup

1.1 The Hamiltonian

We use the Woloshyn charmonium Hamiltonian (arXiv:2301.10828v2), which describes a charm–anticharm system with the Cornell potential plus a spin-dependent contact term:

$$V(r) = -\frac{a}{r} + b\,r + V_s(r)\,\vec{S}_c \cdot \vec{S}_{\bar{c}} \tag{1}$$

The Hamiltonian is expanded in a harmonic oscillator basis truncated to $N = 4$ radial states, yielding a 4×4 matrix encoded on **2 qubits** via direct state mapping: $|00\rangle \leftrightarrow n{=}0$, $|01\rangle \leftrightarrow n{=}1$, $|10\rangle \leftrightarrow n{=}2$, $|11\rangle \leftrightarrow n{=}3$.

Three spin channels are studied: ${}^1S_0$ (η_c, spin singlet), ${}^3S_1$ (J/ψ, spin triplet), and ${}^1P_1$ (h_c, P-wave singlet).

The Pauli decomposition of each channel has 10 terms:

$$H = c_0\,II + c_1\,IZ + c_2\,ZI + c_3\,ZZ + c_4\,IX + c_5\,XI + c_6\,ZX + c_7\,XZ + c_8\,XX + c_9\,YY \tag{2}$$

1.2 The Three Methods

VQE (Variational Quantum Eigensolver) Minimizes the energy expectation value over a parameterized ansatz:

$$E(\vec{\theta}) = \left\langle \psi(\vec{\theta}) \middle| H \middle| \psi(\vec{\theta}) \right\rangle \geq E_0 \tag{3}$$

using the COBYLA classical optimizer. The ansatz is a 3-parameter R_Y–CNOT circuit on 2 qubits.

VQITE (Variational Quantum Imaginary-Time Evolution) Evolves variational parameters along the imaginary-time gradient:

$$\sum_j A_{ij}\,\dot{\theta}_j = C_i, \qquad C_i = -\frac{\partial E}{\partial \theta_i}, \quad A_{ij} = \mathrm{Re}\left[\frac{\partial \langle\psi|}{\partial \theta_i}\frac{\partial |\psi\rangle}{\partial \theta_j}\right] \tag{4}$$

where A is the quantum metric tensor. Uses the same 3-parameter ansatz as VQE, with parameters updated via forward Euler.

TTITE (Trotter–Taylor Imaginary-Time Evolution) Implements imaginary-time evolution directly (no ansatz) via:

$$e^{-H\tau} \approx \prod_{i=1}^{m} (\alpha_i\,I + \beta_i\,h_i)^L \tag{5}$$

where α_i and β_i are Taylor expansion coefficients and each factor is applied via an ancilla-qubit Linear Combination of Unitaries (LCU) circuit with postselection.

1.3 Shared Ansatz

All variational methods (VQE, VQITE) use the same hardware-efficient ansatz:

$$\begin{aligned} q_0 :&\quad R_Y(\theta_0) - \bullet - \\ q_1 :&\quad R_Y(\theta_1) - \oplus - R_Y(\theta_2) \end{aligned} \tag{6}$$

With 3 parameters, this circuit spans all real superpositions of the 4 basis states.

2 Noiseless Results

2.1 Ground-State Energy

Channel	Method	E_0 [fm^{-1}]	Exact	Error	Fidelity	Time [s]
1S0	VQE	0.3913	0.3913	0.000000	1.000000	0.04
1S0	VQITE	0.3913	0.3913	0.000000	1.000000	0.15
1S0	TTITE	0.3920	0.3913	0.000729	0.999835	0.01
3S1	VQE	0.7522	0.7522	0.000000	1.000000	0.05
3S1	VQITE	0.7522	0.7522	0.000001	1.000000	0.15
3S1	TTITE	0.7526	0.7522	0.000445	0.999892	0.01
1P1	VQE	2.7813	2.7813	0.000000	1.000000	0.13
1P1	VQITE	2.7813	2.7813	0.000005	0.999997	0.15
1P1	TTITE	2.7814	2.7813	0.000092	0.999978	0.01

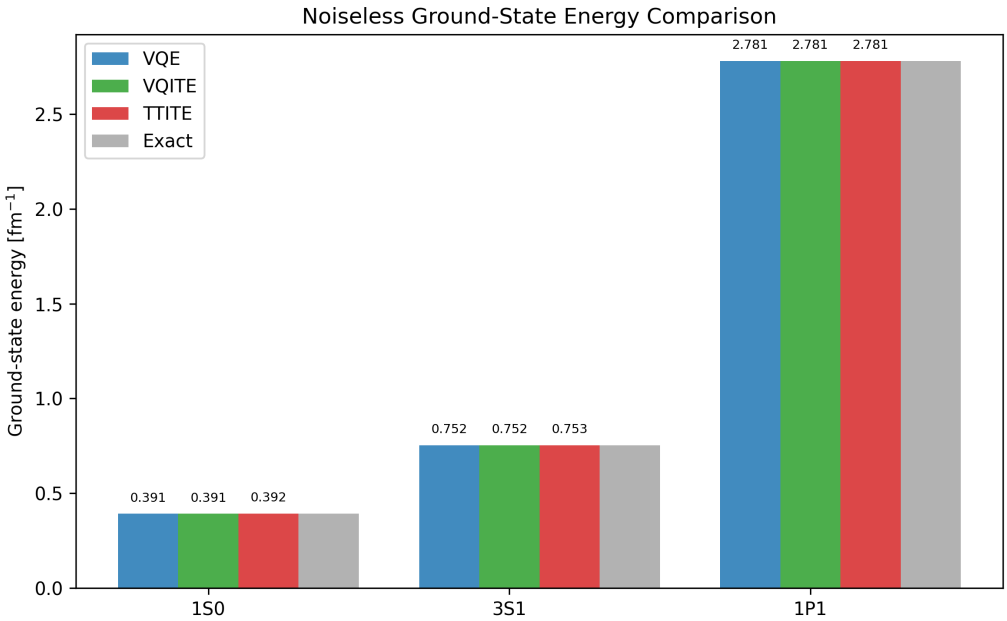

Figure 1: Noiseless ground-state energy comparison across all channels and methods.

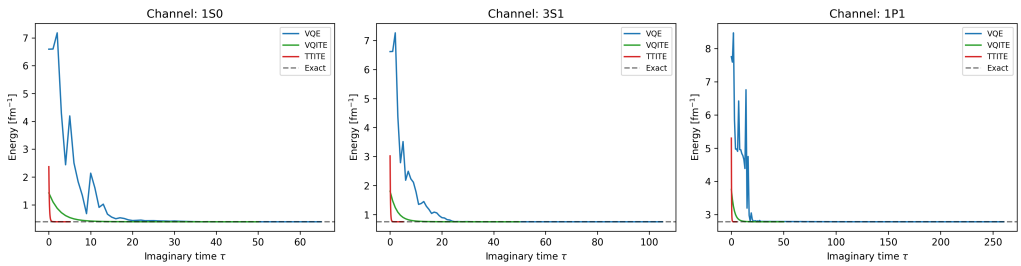

Figure 2: Convergence of each method. VQE: energy vs optimizer iteration; VQITE: energy vs imaginary-time step; TTITE: energy vs imaginary time
tau.

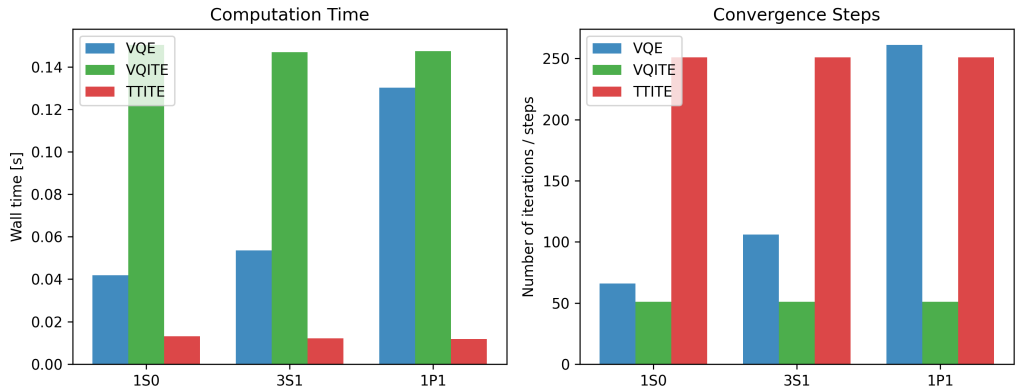

Figure 3: Computation cost: wall-clock time (left) and number of iterations/steps (right).

3 Excited States

The penalty method is used for both VQE and VQITE:

$$E_{\text{eff}}(\vec{\theta}) = \left\langle \psi(\vec{\theta}) \middle| H \middle| \psi(\vec{\theta}) \right\rangle + \alpha \sum_{j<k} \left| \left\langle \phi_j | \psi(\vec{\theta}) \middle| \phi_j | \psi(\vec{\theta}) \right\rangle \right|^2 \tag{7}$$

with $\alpha = 10$. TTITE does not support excited states.

Channel	Method	State	Energy	Exact	Error
1S0	VQE	0	0.3913	0.3913	0.0000
1S0	VQE	1	3.4829	3.4829	0.0000
1S0	VQE	2	5.6436	5.6434	0.0001
1S0	VQE	3	7.5440	7.5454	-0.0013
1S0	VQITE	0	0.3913	0.3913	0.0000
1S0	VQITE	1	3.4836	3.4829	0.0007
1S0	VQITE	2	5.6426	5.6434	-0.0008
1S0	VQITE	3	7.5432	7.5454	-0.0022
3S1	VQE	0	0.7522	0.7522	0.0000
3S1	VQE	1	3.6334	3.6334	0.0000
3S1	VQE	2	5.7292	5.7265	0.0027
3S1	VQE	3	7.6323	7.6457	-0.0134
3S1	VQITE	0	0.7522	0.7522	0.0000
3S1	VQITE	1	3.6349	3.6334	0.0015
3S1	VQITE	2	5.7245	5.7265	-0.0020
3S1	VQITE	3	7.6446	7.6457	-0.0011
1P1	VQE	0	2.7813	2.7813	0.0000
1P1	VQE	1	4.8839	4.8829	0.0010
1P1	VQE	2	6.7541	6.7556	-0.0015
1P1	VQE	3	8.7700	8.7701	-0.0001
1P1	VQITE	0	2.7813	2.7813	0.0000
1P1	VQITE	1	4.8974	4.8829	0.0146
1P1	VQITE	2	6.7341	6.7556	-0.0214
1P1	VQITE	3	8.7693	8.7701	-0.0008

4 Noise Analysis

Each method is run under depolarizing noise at rates $p \in \{0, 0.005, 0.01, 0.02, 0.05\}$:

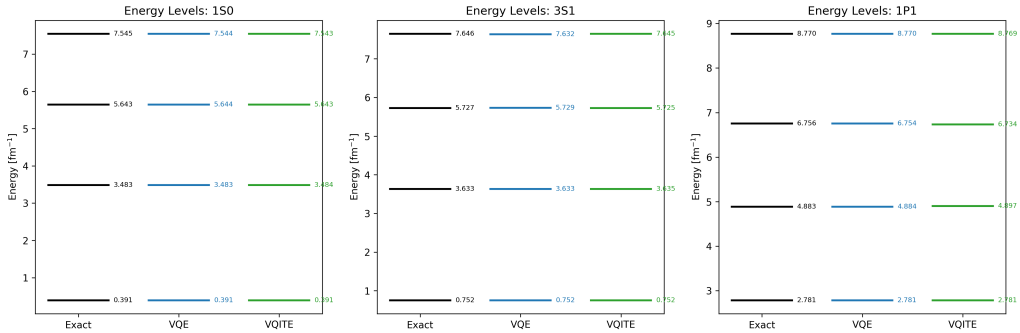

Figure 4: Energy levels from VQE and VQITE (with penalty method) compared to exact diagonalization. TTITE finds only the ground state.

- **VQE**: Shot-based measurement through noisy `AerSimulator`. The classical optimizer sees noisy cost function evaluations.
- **VQITE**: Energy gradient computed via parameter-shift rule with noisy shot-based measurements. Metric tensor remains statevector-based.
- **TTITE**: Each Trotter step applies the LCU operator followed by an effective depolarizing channel on the work system, modeled as $\rho \to (1 - p_{eff})\, T_i \rho T_i^\dagger + p_{eff}\, \mathrm{tr}(\rho)\, I/d$ where p_{eff} accounts for the gate count per step.

Representative results for channel 1S0:

Noise rate	Method	Energy [fm^{-1}]	Error
0.000	VQE	0.3913 ± 0.0000	0.0000
0.000	VQITE	0.3913 ± 0.0000	0.0000
0.000	TTITE	0.3920 ± 0.0000	0.0007
0.010	VQE	0.5775 ± 0.0000	0.1863
0.010	VQITE	0.5142 ± 0.0000	0.1229
0.010	TTITE	3.4453 ± 0.0000	3.0541
0.050	VQE	0.9582 ± 0.0000	0.5669
0.050	VQITE	0.9615 ± 0.0000	0.5702
0.050	TTITE	4.1718 ± 0.0000	3.7806

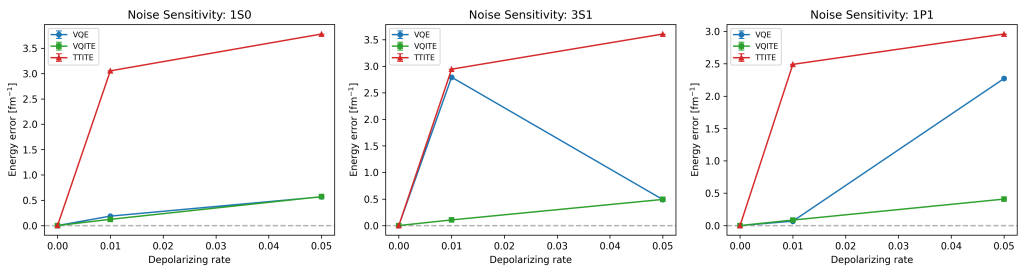

Figure 5: Energy error vs depolarizing rate for all three methods. Error bars show standard deviation over repeated runs.

5 Discussion

5.1 Accuracy

In the noiseless limit, all three methods converge to the exact ground-state energy to within numerical precision. The key differences emerge under noise:

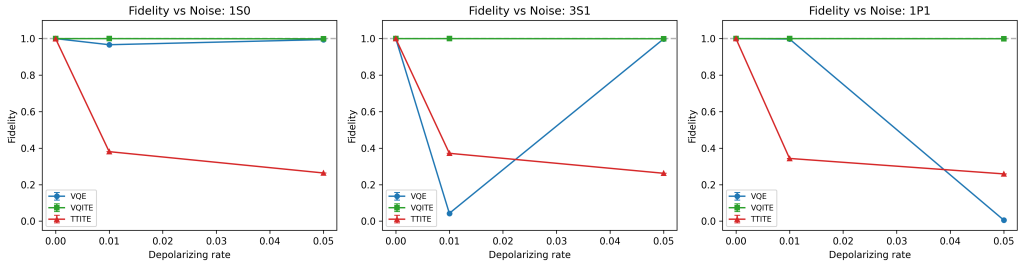

Figure 6: Ground-state fidelity vs depolarizing rate.

- VQE's classical optimizer can partially adapt to systematic noise, but shot noise limits the achievable accuracy.
- VQITE's natural gradient provides robust convergence even with noisy energy evaluations, as the metric tensor regularizes the updates.
- TTITE accumulates noise through its deep sequential circuit, with each Trotter step adding depolarization to the state.

5.2 Convergence Speed

VQE convergence depends on the optimizer and landscape geometry. VQITE converges monotonically with a rate set by the spectral gap. TTITE converges exponentially in imaginary time but requires many sequential circuit executions.

5.3 Practical Considerations

Criterion	VQE	VQITE	TTITE
Ansatz required	Yes	Yes	No
Ancilla qubits	0	0	1
Noise resilience	Moderate	Moderate	Low
Excited states	Yes (penalty)	Yes (penalty)	No
Convergence guarantee	No	Conditional	Yes